

Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction

音频多挑战：在自然人机交互中对语音对话系统进行多轮评估

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Abstract

摘要

End-to-end (E2E) spoken dialogue systems are increasingly replacing cascaded pipelines for voice-based human-AI interaction, processing raw audio directly without intermediate transcription. Existing benchmarks primarily evaluate these models on synthetic speech and single-turn tasks, leaving realistic multi-turn conversational ability underexplored. We introduce Audio MultiChallenge, an open-source benchmark to evaluate E2E spoken dialogue systems under natural multi-turn interaction patterns. Building on the text-based MultiChallenge framework, which evaluates Inference Memory, Instruction Retention, and Self Coherence, we introduce a new axis Voice Editing that tests robustness to mid-utterance speech repairs and backtracking. We further augment each axis to the audio modality, such as introducing Audio-Cue challenges for Inference Memory that require recalling ambient sounds and paralinguistic signals beyond semantic content. We curate 452 conversations from 47 speakers with 1,712 instance-specific rubrics through a hybrid audio-native agentic and human-in-the-loop pipeline that exposes model failures at scale while preserving natural disfluencies found in unscripted human speech. Our evaluation of proprietary and open-source models reveals that even frontier models struggle on our benchmark, with Gemini 3 Pro Preview (Thinking), our highest-performing model achieving a 54.65% pass rate. Error analysis shows that models fail most often on our new axes and that Self Coherence degrades with longer audio context. These failures reflect difficulty of tracking edits, audio cues, and long-range context in natural spoken dialogue. Audio MultiChallenge provides a reproducible testbed to quantify them and drive improvements in audio-native multi-turn interaction capability.

端到端 (E2E) 语音对话系统正越来越多地取代基于级联流水线的语音人机交互方式，直接处理原始音频而无需中间的转录过程。现有的基准测试主要在合成语音和单轮任务上评估这些模型，使得现实中的多轮对话能力研究仍显不足。我们引入了 Audio MultiChallenge，这是一个开源基准测试，用于在自然的多轮交互模式下评估端到端语音对话系统。基于文本型 MultiChallenge 框架，该框架评估推理记忆、指令

保留和自我连贯性，我们引入了一个新的维度“语音编辑”，用于测试对语句中途语音修正和回溯的鲁棒性。我们进一步将每个维度扩展到音频模态，例如引入音频线索挑战来测试推理记忆，这些挑战要求模型回忆超出语义内容的环境声音和副语言信号。我们通过混合音频原生代理和人类参与的流程，收集了 47 位说话人的 452 次对话，并使用 1,712 个特定实例的评分标准，该流程在大规模暴露模型失败的同时保留了非脚本人类语音中自然的不流畅性。我们对专有和开源模型的评估表明，即使是前沿模型也难以在我们的基准测试中表现良好，我们的表现最好的模型 Gemini 3 Pro Preview (Thinking) 达到了 54.65% 的通过率。错误分析表明，模型在我们新引入的维度上失败最为频繁，并且随着音频上下文的延长，自我连贯性会下降。这些失败反映了在自然语音对话中跟踪编辑、音频线索和长距离上下文的困难。Audio MultiChallenge 提供了一个可重复的测试平台，用于量化这些失败并推动音频原生多轮交互能力的改进。

1. Introduction

1. 引言

Spoken dialogue systems [1] are evolving from cascaded pipelines comprising Automatic Speech Recognition (ASR) [2], text-based Large Language Models (LLMs) [3], and Text-to-Speech (TTS) systems [4] to native end-to-end (E2E) architectures that map input speech directly to conversational responses without an intermediate transcription step [5–7]. Speech-to-speech (S2S) models and broader audio language models (Audio LMs) process raw or minimally processed acoustic features and generate spoken or textual outputs, often operating over shared latent representations or learned discrete speech units rather than relying solely on text tokens [5, 8–11]. This integration can reduce response latency and mitigate error propagation, while better preserving paralinguistic and acoustic information that is often lost in cascaded systems [12, 13].

语音对话系统 [1] 正在从由自动语音识别 (ASR) [2]、基于文本的大语言模型 (LLMs) [3] 和文本到语音 (TTS) 系统 [4] 组成的级联流水线，演进为原生的端到端 (E2E) 架构，该架构可直接将输入语音映射为对话响应，而无需中间的转录步骤 [5–7]。语音到语音 (S2S) 模型和更广泛的音频语言模型 (音频 LMs) 处理原始或经过最少处理的声学特征，并生成语音或文本输出，通常基于共享的潜在表示或学习到的离散语音单元，而不是仅依赖文本标记 [5, 8–11]。这种集成可以减少响应延迟并减轻错误传播，同时更好地保留通常在级联系统中丢失的副语言和声学信息 [12, 13]。

As E2E systems move toward real-world user-facing interaction, their evaluation must reflect the demands of extended, spontaneous dialogue rather than isolated prompt-response pairs or short, synthetic TTS inputs common in existing benchmarks [14–16]. Evidence from text-only LLMs shows that strong single-turn performance does not guarantee robust multi-turn behavior [17–19]. These weaknesses are amplified in the audio modality. Audio is temporally continuous and often represented as high-frame-rate token streams, which rapidly consume context and make long-horizon state tracking harder than in text [9, 12]. Consequently, while existing Audio LM benchmarks suggest strong performance on short, clean, or synthetic inputs, robustness in realistic multi-turn settings remains understudied. Even recent multi-turn efforts are limited to short dialogues [20], leaving unclear whether systems

随着端到端系统向真实世界面向用户的交互发展，其评估必须反映扩展对话和自发对话的需求，而不仅仅是孤立的提示-响应对或现有基准中常见的短且合成的 TTS 输入 [14–16]。文本仅的 LLM 证据表明，强大的单轮性能并不能保证稳健的多轮行为 [17–19]。这些弱点在音频模态中被放大。音频是时间连续的，通常表示为高帧率的标记流，这会迅速消耗上下文，使得长时域状态跟踪比文本更困难 [9, 12]。因此，尽管现有的音频 LM 基准表明在短、干净或合成输入上表现强劲，但在现实多轮设置中的鲁棒性仍研究不足。即使最近的多轮研究也仅限于短对话 [20]，使得系统是否能够在更复杂的多轮场景中保持鲁棒性仍不清楚。

Inference Memory

推理记忆

I. Semantic Memory

I. 语义记忆



Figure 1: images/7f21d4f9d4d7ad4941e5efd895ebfc9b0775b6285f63555d6d3c6953dfe9e789.jpg

I'm brainstorming ideas for my next romance novel, and I think I want to open it with a couple arguing on a ferry. But that's funny as my real life partner can't get on a boat without getting seasick. Anyway, can you help me...

我正在为我的下一部长篇爱情小说构思创意，我想以一对情侣在渡轮上争吵作为开头。不过这很有趣，因为我的现实中的伴侣一坐船就会晕船。不管怎样，你能帮我吗.....

Of course. You could use the ferry itself to build the tension. Maybe the wind is sharp...

当然。你可以利用渡轮本身来制造紧张感。也许风很猛烈.....



Figure 2: images/b740ecff7b7fc734c94b1697ba5d3e83feba35b304e6e0eb0f978266db145c7.jpg



Figure 3: images/c24f0c6d8f1f661ba0b6e4d58b8970639062e8cbcc95ca2e06be2d48d8da7041.jpg

That's great. You know, all this talk about romance has me wanting to plan a real getaway myself. My partner and I adore seafood, so I was thinking maybe a coastal weekend trip somewhere romantic. What are some fun things we could do by the water?

这很好。你知道吗，所有关于浪漫的谈论让我也想自己计划一次真正的旅行。我和我的伴侣都非常喜欢海鲜，所以我正想着或许可以去一个浪漫的地方度过一个海边周末。在水边我们能做些什么有趣的事情呢？

II. Audio-Cue Memory

II. 音频提示记忆



Figure 4: images/c70e158b38868d6287510541d71cd7fd41ad4d0d27d009baad435c32bb90ddab.jpg

I've been looking for the last great fairy fountain in Breath of the Wild for like an hour now, do you have any tips to find it?

我一直在寻找《塞尔达传说：荒野之息》中最后一个伟大的仙女泉，已经找了一个小时了，你有任何找到它的建议吗？

The last one is usually in the Gerudo Desert. Head to the far southwest corner of the map...

最后一个通常位于盖鲁多沙漠。前往地图的西南角...



Figure 5: images/c81075118ad85a0fa4d9d78062fba4900675ef8f89e7ee9f66f4336e5727f752.jpg



Figure 6: images/a692ad3b48506d01b803238850cbabbbcf7ab82fc4ff6e0ab89e10d1beab148a.jpg

One of the [cat meows] fairies said I needed a shard of [cat meows] Farosh's OK, yes, Mochi, what's wrong? OK. All right. Thank you, Mochi.

其中一个 [cat meows] 精灵说我需要一块 [cat meows] 法罗斯的碎片，好的，是的，莫奇，怎么了？好的。好吧。谢谢，莫奇。

Farosh is the giant green spirit dragon that flies around Hyrule. To get a shard...

法罗斯是围绕海拉鲁飞行的巨大绿色精神龙。要获得一块碎片...



Figure 7: images/63a70184de031aab1bc156a93cf4ae9c330238e93db5b1dbc68f15f8f42806bb.jpg



Figure 8: images/c77b0ee892176125459e38845e7946796b06c3604acd7937207625108f8a4e75.jpg



Figure 9: images/d3d722fd7cb0289e97d639920b935f01992ba1e138956e8d00b449b79b64bb68.jpg

Does the same thing apply to [interruption] huh? No, I'll be there in just a minute, um, does the same thing apply to Tears of the Kingdom

是否同样适用于 [中断] 呢? 嗯? 不, 我马上就到, 嗯, 是否同样适用于《王国之泪》?

It's very similar. You can still shoot the dragons...

它非常相似。你仍然可以射杀龙.....



Figure 10: images/5c6945541b5ff332ebe9089782a5bc7e1fa3de88f4f7f4cd97fd93a464af5d2b.jpg



Figure 1: Samples from Audio MultiChallenge. Highlighted text shows key interaction patterns per axis.

图 1: Audio MultiChallenge 的示例。突出显示的文本展示了每个轴的关键交互模式。

Does Mochi need a FVRCP or DHPP vaccine?

狗狗是否需要接种 FVRCP 或 DHPP 疫苗?

Instruction Retention

指令保留



Figure 11: images/42827f243258547d8442e4a3d5bfd87372878879424776145b1509dd79ccd2b0.jpg

Can you give me suggestions about what grows well here in St. Pete, Florida? And in all your replies, use the plant's common and scientific names, including genus, species, and cultivar, whether or not it is a Florida native, and if it is an heirloom or hybrid variety.

你能给我一些关于在佛罗里达州圣彼得斯堡地区生长良好的植物的建议吗? 在你所有的回复中, 请使用植物的通用名称和科学名称, 包括属、种和栽培品种, 无论它是否是佛罗里达州的原产植物, 以及它是否是传统品种或杂交品种。

For your garden, a great choice is the Everglades Tomato. That's *Solanum pimpinellifolium*, the Everglades' cultivar. It's a wonderful Florida native heirloom that thrives in the heat.

对于你的花园，一个很好的选择是“佛罗里达番茄”（Everglades Tomato）。这是一种名为 *Solanum pimpinellifolium* 的品种，是佛罗里达州的本土传统品种。它在炎热的气候中生长旺盛。



Figure 12: images/c8c92efabc56d98189129353e8f702c8328ac6f0bd4791553a3f0a6c8aa18382.jpg



Figure 13: images/badc91f27a5ea0cf5613bf2d92e32196e891cc50716e1261d435e8ec028e4da0.jpg

Wow, those sound really good. Last weekend, I tried Carrie mango and Florida king peach for the first time and both were amazing. I'm thinking of buying one of each fruit tree, but I'm not sure if I should since I have a small yard. How big do those grow?

哇，这些听起来真的很棒。上周末，我第一次尝试了卡丽芒果和佛罗里达王 peach，两者都非常美味。我打算各买一棵这种果树，但我不确定是否应该这样做，因为我有一个小院子。这些树长大后会有多大？

Self Coherence

自我一致性



Figure 14: images/1febabcf70feaa8bb24d8e4c248575b9c69ba38a03dfa6218f281cd7cd16da79.jpg

I'm studying for my med exam and I need a refresher on some things. Um, what bones work together in the shoulder for rotation?

我正在为医学考试做复习，需要一些知识点的回顾。嗯，肩部有哪些骨骼参与旋转？

The shoulder involves three main bones. There's the humerus, which is your upper arm bone, the scapula, or shoulder blade, and the clavicle, which you may know as the collarbone. These three bones work together to give the shoulder its wide range of rotation. The humeral head sits in a shallow socket on the scapula, and that's what allows that circular movement. The clavicle helps stabilize it all.

肩部涉及三块主要的骨头。分别是肱骨（上臂骨）、肩胛骨（肩胛骨）和锁骨（胸骨）。这三块骨头协同工作，使肩部具有广泛的旋转范围。肱骨头位于肩胛骨的一个浅窝中，正是这种结构使得肩部能够进行圆周运动。锁骨则帮助稳定整个结构。



Figure 15: images/02023f4c15905a89f597c7e771dde2dbcff267ce0341b7e89b177b570f40c9e0.jpg



Figure 16: images/70606f5e2495bf598bee119eefb2de21c52d8c0642fcccfe9e30db4908bd10e1.jpg



Figure 17: images/6db083afeb58d9a285164d966b6285f26bb9496796ec19b76dfae9c64429e61f.jpg

That's great, I'll have it checked out. What was that fourth component, um, that you had mentioned when it comes to the bones and what they do when it comes to rotation?

这很好，我会去查看一下。你说的第四部分是什么呢，就是关于骨骼以及它们在旋转方面的作用？

Voice Editing

语音编辑

I. Mid-Utterance Edits

I. 语中编辑

II. Previous-Turn Edits

II. 上一回合的编辑



Figure 18: images/6fc651ea37f5eef2cafc8329f77ae8ba3cde3ac7a7412982d954f6858748fc41.jpg

Can you help me plan my landscape? I've got some big ideas and I think I need some help.

你能帮我规划我的景观吗？我有一些大想法，我认为我需要一些帮助。

A great first step is to think about how you want to use the space, like for relaxing or...

一个很好的第一步是考虑你想要如何使用这个空间，比如用来放松或.....



Figure 19: images/14ec200eba22f29590372d167e5126dc92fa852d7fa5247ffaa1ba09cd36444d.jpg



Figure 20: images/e59e31bb03a7778fbca62602d1b710f67b5361865c04b1dcd5d34524670be0ce.jpg

Definitely relaxing. I want to add a deck, a hot tub. Let's start with two weeping willow trees. No, make that three weeping willow trees.

肯定很放松。我想加一个露台和一个热水浴池。我们先从两棵垂柳树开始。不，改成三棵垂柳树吧。

That sounds like a wonderfully relaxing oasis. One thing to keep in mind with weeping willows...

那听起来像是一个美妙的放松天堂。关于垂柳.....有一点需要注意的是.....



Figure 21: images/d82d203cca6d2a9a5229dbada242ed2c41a69359d41e1d3f2da39db3e09b1d2b.jpg



Figure 22: images/b5f21a3fdc63caf4cda71d643e5e8abb2f2e6cbc0399549db117374fc39c4134.jpg

Okay that's good to know, thank you. let's make it two weeping willow trees, four yuccas, hmm no six. Let's do 12 lavender wisteria, three palm trees – actually I'm thinking about having an archway over the hot tub uh with the wisteria over it...

好的，知道了，谢谢。我们来种两棵 weeping willow 树，四棵 yucca，嗯，不，六棵。我们来做十二棵薰衣草的紫藤，三棵棕榈树——实际上我是在考虑在热水浴池上方建一个拱门，用紫藤覆盖在上面.....

An archway with wisteria sounds gorgeous. Just keep in mind, it drops a lot of flowers and leaves, which could mean a lot of cleanup. You could get a similar feel, with less maintenance, by...

一座爬满紫藤的拱门看起来非常漂亮。不过请注意，它会掉落很多花朵和叶子，这意味着需要做很多清洁工作。你也可以选择一种类似的感觉，但维护起来更轻松的方式，比如.....



Figure 23: images/ddaa90e84a046e0a21e548a7ffaa8b52a5097ed383a9b5effdaccdd6fa5fecb19.jpg



Figure 24: images/44102281f77f4437ad81bb2c799e4cb33c8a2f8420bb1a97ceaa058b4b6b7239.jpg

Yeah, let's go with the trellis. That sounds good. Let's add river rock, one ton, make it two tons. Add a pond with 12 koi fish. Let's make it six palm trees. Add six more yuccas, hmm, make the pond larger with 24 koi fish.

是的，我们用篱笆吧。听起来不错。再加些河岩，一吨，做成两吨。添加一个池塘，里面有 12 条锦鲤。我们做六棵棕榈树。再加六棵尤加利叶，嗯，把池塘做大一些，里面有 24 条锦鲤。

That sounds incredible. A pond that can support 24 koi will need to be quite large. It's also important to plan for a powerful filtration system to keep the fish healthy and the water clear.

这听起来令人难以置信。一个能够支持 24 尾锦鲤的池塘需要相当大。另外，还需要计划安装一个强大的过滤系统，以保持鱼的健康和水质清澈。



Figure 25: images/6cd255318eb4a13a63e28f7542006f9582fbdafd5211e64d73cefe401f7e10d2.jpg



Figure 26: images/cd6241653c8dc875d05a1f6af92e89cf9de747a4d0ae07162b64827b7f2119d.jpg

So, how large is quite large? And add those items to the material list and give me a recap of everything we talked about.

所以，相当大的规模有多大呢？请将这些项目添加到材料清单中，并给我一个我们讨论内容的总结。

can handle key interaction patterns such as maintaining self-consistency, following user constraints, leveraging audio cues over long contexts, and recovering from user repairs like mid-utterance self-corrections and barge-ins.

可以处理诸如保持自治性、遵循用户约束、在长上下文中利用音频线索，以及从用户修复（如中途发言的自我纠正和打断）中恢复等关键交互模式。

To rigorously evaluate these dimensions, we open-source Audio MultiChallenge, an extension of the text-based MultiChallenge evaluation framework [17] to the audio modality through three key contributions:

为了严格评估这些维度，我们开源了 Audio MultiChallenge，这是基于文本的 MultiChallenge 评估框架 [17] 的扩展，通过以下三个关键贡献，将其扩展到了音频模式：

- Audio-native axes and tasks. We introduce a novel Voice Editing axis that evaluates robustness to mid-utterance speech repairs and backtracking, capturing natural human behaviors such as self-corrections that are common in spontaneous dialogue. We also add Audio-Cue Inference Memory tasks that require recalling ambient sounds or paralinguistic signals from previous turns, rather than only semantic content, and adapt the remaining

Instruction Retention and Self Coherence axes to better reflect the complexities of voice interactions.

- Hybrid data generation for scalability and diversity. We introduce a data-generation pipeline that combines multi-agent synthetic component with a human-in-the-loop process to produce our final multi-turn conversation prompts. Rather than providing human contributors with verbatim scripts, we provide high-level blueprints and interaction goals derived from a scalable agentic loop designed to elicit diverse failure modes. These blueprints encourage improvisational interactions that surface realistic failures, making our dataset sufficiently adversarial while remaining entirely unscripted and preserving paralinguistics and natural human disfluencies.
- Fine-grained rubric-based evaluation. We replace coarse binary success metrics with 1,712 atomic, instance-specific rubrics across the dataset that decompose each ideal response into critical sub-requirements, and our LLM-as-a-judge method using these rubrics achieves 93% agreement rate with human evaluators. While satisfying all corresponding rubrics is required to pass a task, individual rubrics provide high-resolution attribution of common model failures.
- 音频原生的轴与任务。我们引入了一个新颖的语音编辑轴，用于评估对话中语音修复和回溯的鲁棒性，捕捉自然人类行为，例如自发对话中常见的自我纠正。我们还增加了音频线索推理记忆任务，这些任务要求从之前的对话回合中回忆环境声音或副语言信号，而不仅仅是语义内容，并调整了剩余的指令保留和自我连贯性轴，以更好地反映语音交互的复杂性。
- 混合数据生成以实现可扩展性和多样性。我们引入了一个数据生成流水线，结合多智能体合成组件与人类参与流程，以生成最终的多轮对话提示。我们不向人类贡献者提供逐字脚本，而是提供基于可扩展智能体循环设计的高层次蓝图和交互目标，这些蓝图旨在激发多样化的失败模式。这些蓝图鼓励即兴互动，揭示现实的失败情况，使我们的数据集在保持完全非脚本化的同时，具备充分的对抗性，并保留副语言特征和自然的人类不流畅性。
- 细粒度基于评分标准的评估。我们用 1,712 个原子化、实例特定的评分标准替代了粗略的二元成功指标，这些评分标准贯穿整个数据集，将每个理想响应分解为关键的子要求，我们的使用这些评分标准的 LLM 作为评判方法与人类评估者达成 93% 的一致率。虽然满足所有相应的评分标准是通过任务的必要条件，但个别评分标准提供了对常见模型失败的高分辨率归因。

Audio MultiChallenge comprises 452 multi-turn conversations from 47 unique speakers, summing up to 15 hours

Audio MultiChallenge 包含来自 47 位独特说话者的 452 次多轮对话，总计 15 小时

of user audio recorded without post-processing to retain ambient acoustic variability.

未经后期处理的用户音频记录，以保留环境声学的变异性。

Our analysis of Audio MultiChallenge reveals substantial gaps in current E2E capabilities across frontier models. Even the highest-performing model, Gemini 3 Pro Preview (Thinking), achieves only 54.65% pass rate, with other model families, especially open-source ones lagging considerably behind. We find that Self Coherence degrades consistently as cumulative user audio duration increases, dropping from 33.3% on shorter tasks to 20.0% on tasks with 3–5 minutes of audio. Models also struggle to recall paralinguistic and ambient signals from previous turns, scoring 36.5% lower on Audio-Cue Inference Memory than on semantic memory tasks. Voice Editing proves to be the most challenging axis, where behaviors such as mid-utterance self-corrections and prior-turn edits frequently trigger failures that are largely absent in cleaner text baselines. Together, our results suggest the need for audio-native training and evaluation that target realistic multi-turn spoken interaction.

我们对 Audio MultiChallenge 的分析揭示了当前前沿模型在端到端能力方面存在显著的不足。即使表现最好的模型 Gemini 3 Pro Preview (Thinking)，也仅实现了 54.65% 的通过率，其他模型家族，尤其是开源模型，表现明显落后。我们发现，随着累计用户音频时长的增加，Self Coherence 会持续下降，从较短任务中的 33.3% 下降到包含 3-5 分钟音频的任务中的 20.0%。模型在回忆会话中非语言信号和环境信号方面也存在困难，在音频线索推理记忆任务上的得分比语义记忆任务低 36.5%。语音编辑被证明是最具挑战性的方面，其中诸如中途自我修正和先前回合的编辑等行为经常引发失败，而这些失败在干净文本基线中几乎不存在。总体而言，我们的结果表明，需要针对现实多轮口语交互的音频原生训练和评估。

2. Audio MultiChallenge

2. 音频多挑战

In Audio MultiChallenge, we focus on four core axes designed to test the multi-turn capabilities E2E spoken dialogue systems: Inference Memory, Instruction Retention, Self Coherence, and Voice Editing. The fourth axis, Voice Editing, is a novel addition specific to the speech domain. We introduce this axis to address a fundamental difference between modalities. Unlike text inputs where errors are typically backspaced prior to submission, speech is a continuous stream where the editing process is audible. This necessitates that models dynamically filter retracted information to isolate the user's final intent. Collectively, these categories represent practical, real-world conversational hurdles that remain challenging for current frontier models and have historically been evaluated primarily on single-turn tasks. For each category, we release a set of human-recorded, non-scripted test examples curated according to the specific axis definition.

在 Audio MultiChallenge 中，我们专注于四个核心轴，旨在测试端到端语音对话系统在多轮交互中的能力：推理记忆、指令保持、自我一致性以及语音编辑。其中，第四轴“语音编辑”是专门针对语音领域的创新性新增内容。我们引入这一轴以解决不同模态之间的一个基本差异。与文本输入中通常在提交前通过退格键纠正错误不同，语音是一种连续的流，编辑过程是可听的。这要求模型能够动态过滤撤回的信息，以隔离用户的最终意图。总体而言，这些类别代表了实际、现实中的对话挑战，目前前沿模型在这些方面仍面临困难，并且历史上主要在单轮任务中进行评估。对于每个类别，我们发布了一组由人类录制、非脚本化的测试示例，并根据特定轴的定义进行筛选。

We explicitly distinguish Audio MultiChallenge from text-based MultiChallenge to address the unique requirements of spoken interaction. Although MultiChallenge offers a strong foundation, simply applying

我们明确区分 Audio MultiChallenge 与基于文本的 MultiChallenge，以应对口语交互的独特需求。尽管 MultiChallenge 提供了坚实的基础，但仅仅应用

Conversations	452
Inference Memory	132
Instruction Retention	120
Self Coherence	117
Voice Editing	83
Turn Count	3-8
Rubrics	1,712
Speakers	47
User Audio Duration	14.99 h
Sampling Rate	48 kHz
Language	English

对话	452
推理记忆	132
指令保留	120
自我一致性	117
语音编辑	83
计数器	3-8
评分标准	1,712
演讲者	47
用户音频时长	14.99 美元
采样率	48 kHz
语言	中文

Table 1: Dataset Statistics

表 1: 数据集统计信息

TTS to the existing text dataset is insufficient. For instance, many text-based interactions such as writing code or formatting text do not naturally translate to voice. Audio MultiChallenge instead captures the inherent complexity of spontaneous human speech by curating real recordings, which unlike synthetic or scripted data, include natural disfluencies, backtracking, and third-party interruptions or cross-talk. Our approach allows us to evaluate model robustness in terms of semantic understanding as well as against real-world acoustics, varying accents, and complex paralinguistic cues which text-derived benchmarks fail to capture. We define our axes in detail below.

将 TTS 应用于现有的文本数据集是不够的。例如，许多基于文本的交互，如编写代码或格式化文本，并不自然地转化为语音。Audio MultiChallenge 通过精选真实的录音来捕捉自发人类语音的固有复杂性，这些录音不同于合成或脚本数据，包含了自然的停顿、回溯以及第三方干扰或交叉交谈等元素。我们的方法使我们能够从语义理解以及对抗真实世界的声音、不同的口音和复杂的非语言线索等方面评估模型的鲁棒性，而这些是基于文本的基准测试所无法捕捉的。我们将在下面详细定义我们的评估维度。

2.1 Axes

2.1 坐标轴

Inference Memory Inference Memory evaluates the model’s capacity to recall and synthesize scattered user-specified details from previous turns that are implicitly required to satisfy the user’s final request. Unlike direct queries (e.g., “What did I say earlier?”), these tasks demand that the model proactively reallocate attention to prior context to inform its reasoning. For example, a standard Semantic test might require remembering a stated nut allergy when generating a dessert recipe. Audio MultiChallenge extends this definition to include Audio-Cue Gated Inference Memory. In these scenarios, the critical context to be retained is not explicitly spoken, but is embedded in the acoustic environment or paralinguistic signals. An Audio-Cue gated test might feature the background sound of heavy rain in an initial turn, implicitly constraining a later request for an outfit recommendation. This evaluates the model’s ability to retain and integrate both semantic content and acoustic cues over a multi-turn conversation.

推理记忆推理记忆评估模型回忆并综合之前回合中用户指定的零散细节的能力，这些细节隐含地需要满足用户的最终请求。与直接查询（例如，“我之前说了什么？”）不同，这些任务要求模型主动将注意力重新分配到先前的上下文中以指导其推理。例如，一个标准的语义测试可能需要在生成甜点食谱时记住用户提到的坚果过敏情况。音频多挑战测试将这一定义扩展到包括音频提示门控推理记忆。在这些情境中，需要

保留的关键上下文并未明确说出，而是嵌入在声音环境或副语言信号中。一个音频提示门控测试可能在初始回合中包含大雨的背景音，隐含地限制了后续对服装推荐的请求。这评估了模型在多轮对话中保留和整合语义内容和声音提示的能力。

Instruction Retention Instruction Retention measures a model’s ability to follow explicit user instructions consistently across an entire multi-turn conversation. In Audio MultiChallenge, we broaden the instruction space beyond simple stylistic or structural constraints, to stacked and conditional rule sets. These include multi-constraint directives and trigger-based behaviors, for example changing the response format only after a specific phrase is spoken. We also broaden conversation topics here to include more realistic voice applications such as debate and

指令保留（Instruction Retention）衡量模型在多轮对话中持续一致地遵循明确用户指令的能力。在 Audio MultiChallenge 中，我们将指令空间扩展到超越简单的风格或结构限制，包括堆叠式和条件规则集。这些规则集包括多约束指令和基于触发的行为，例如在特定短语被说出后才更改响应格式。我们还在此处扩展了对话主题，以包含更真实的语音应用，例如辩论和

roleplay. This axis evaluates whether an Audio LM can maintain and correctly apply precise, evolving instructions over long dialogue histories, a failure mode that is frequently observed in speech interactions [21, 22].

角色扮演。该指标评估音频语言模型是否能够在长对话历史中保持并正确应用精确且不断变化的指令，这是一种在语音交互中经常观察到的失败模式 [21, 22]。

Self Coherence Self Coherence evaluates the model’s ability to maintain internal consistency regarding factual assertions, opinions, and established personas throughout a dialogue. A common failure mode in frontier models is contradicting its prior correct responses to align with a user who questions or challenges them. For Audio MultiChallenge, we refine this category by excluding contrived, adversarial red-teaming prompts in favor of realistic conversational scenarios. While adaptive reasoning is desirable in light of new evidence, we specifically target unwarranted contradictions where the model invalidates its own previous statements without justification.

自我一致性（Self Coherence）评估模型在对话中维持事实陈述、观点和既定人设方面的一致性能力。在前沿模型中，一种常见的失败模式是，为了迎合质疑或挑战它们的用户，而与之前的正确回答产生矛盾。对于 Audio MultiChallenge，我们通过排除人为设计的对抗性红队提示，转而采用更现实的对话场景来细化这一类别。虽然在面对新证据时具备适应性推理是可取的，但我们特别关注那些无端的矛盾，即模型在没有合理依据的情况下否定自己之前的陈述。

Voice Editing Voice Editing is a novel axis introduced specifically to mimic speech interactions with a personal assistant. We define this axis as the model’s ability to recognize and apply immediate, spontaneous speech repairs, such as mid-utterance self-

语音编辑语音编辑是一种新颖的轴线，专门引入以模拟与个人助理的语音交互。我们定义这一轴线为模型能够识别并应用即时、自发的语音修正能力，例如在句子中间进行自我修正

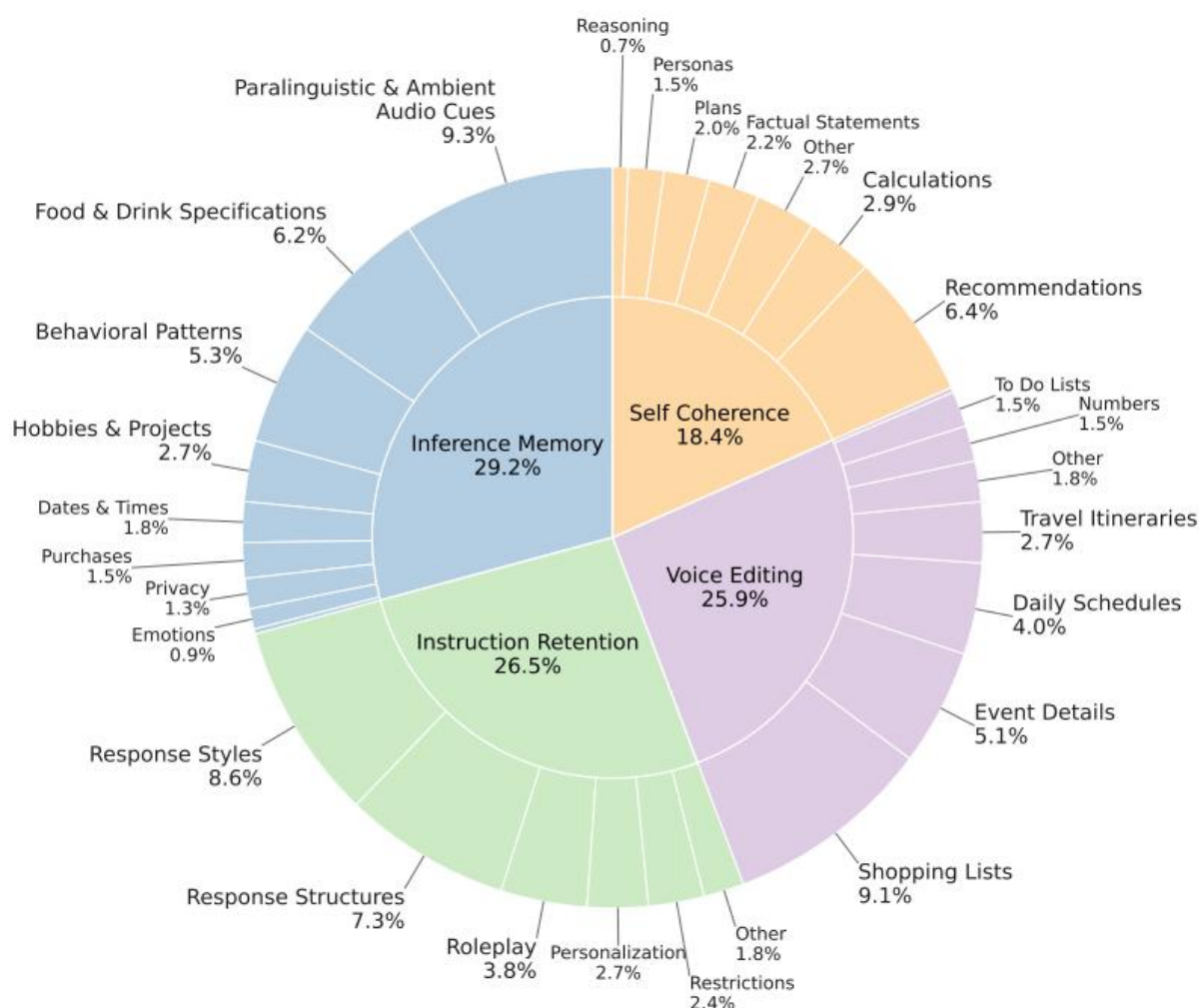


Figure 27: images/d37d022b2651182f5faf4dc34ccb756e62b98106829e874115c5b051a3ddf08e.jpg

Figure 2: Topic Distribution. Detailed breakdown in Appendix A.7

图 2：主题分布。详细分解见附录 A.7

corrections or implemented planned edits that span multiple turns. This axis targets a unique challenge for audio-native models. While LLMs are trained on refined prompts where errors are backspaced or redrafted prior to submission, speech is a continuous stream where the editing process is audible. Consequently, models must dynamically filter out retracted content (e.g., “Let’s make it four yuccas, hmm no six.”) and resolve implicit contradictions that arise within consecutive turns. Evaluation here focuses on the model’s robustness to these natural disfluencies and its capacity to discern the user’s final intent amidst the messiness of spoken interaction.

修正或实施的计划编辑跨越多个回合。这一轴针对音频原生模型的独特挑战。虽然大型语言模型是在经过润色的提示上进行训练的，其中错误会在提交前通过退格或重写进行修正，但语音是一个连续的流，编辑过程是可听见的。因此，模型必须动态地过滤掉撤回的内容（例如，“让我们定为四个尤加利，嗯不，六个。”），并解决连续回合中出现的隐含矛盾。在这里的评估重点在于模型对这些自然不流畅性的鲁棒性，以及其在口语交互的混乱中辨别用户最终意图的能力。

2.2 Dataset Curation

2.2 数据集构建

Producing naturalistic, diverse, and challenging test examples for Audio MultiChallenge is resource intensive. We optimize our dataset curation using a two-stage hybrid approach that combines (i) a multi-agent pipeline to synthetically explore Audio LM failure modes at scale with a (ii) human-in-the-loop stage that translates these failure blueprints into realistic conversations and corresponding evaluation rubrics. An overview is shown in Figure 3, and we describe each step below.

为 Audio MultiChallenge 生成自然、多样且具有挑战性的测试示例需要大量资源。我们采用一种两阶段的混合方法来优化数据集的构建，该方法结合了 (i) 一个多代理管道，以大规模合成探索音频语言模型的失败模式，以及 (ii) 一个包含人类的阶段，将这些失败蓝图转化为真实的对话并对应评估标准。图 3 展示了总体概述，下面我们将逐一描述每个步骤。

Generating Synthetic Audio LM Failures We implement an automated multi-agent framework to generate synthetic conversations that elicit failures along a targeted axis. This approach follows the synthetic generation methodology of MultiChallenge, adapted to the audio modality.

生成合成音频 LM 故障我们实现了一个自动化的多代理框架，以生成合成对话，从而在特定轴线上引发故障。这种方法遵循了 MultiChallenge 的合成生成方法，并将其适应于音频模态。

As shown in Figure 3, the process begins by sampling a target Evaluation Axis A (e.g. Inference Memory) along with a seed topic T (e.g. DietaryRestrictions) and persona P (e.g. Content creator) from curated taxonomies. A Planner Agent (text LLM) then devises an initial conversation strategy aimed at triggering a specific failure mode while adhering to the sampled topic. Based on this strategy, a Tester Agent (text LLM) generates initial user prompt text, which is converted to speech using TTS to simulate audio input. This audio is then fed to the Target Agent (Audio LM). The Planner Agent evaluates the Target Agent’s output to detect failure. If the model succeeds, the Planner updates the strategy to introduce higher complexity in the subsequent turn. This cycle repeats until a failure is detected or a maximum turn count is reached. Once a failure is detected, the Planner Agent analyzes the history H and extracts the core logic or trick that caused the model to fail along the axis. This is synthesized into a concise Human-friendly Blueprint B_{final} , serving as a high-level strategic guide for human contributors. In this study, we use gpt-4o-mini-tts by OpenAI as the TTS system, randomly sample between GPT 4o Audio Preview

如图 3 所示，该过程首先从经过整理的分类法中采样一个目标评估轴 A （例如推理记忆），以及一个种子话题 T （例如饮食限制）和一个角色 P （例如内容创作者）。然后，一个规划代理（文本 LLM）会制定一个初始对话策略，旨在触发特定的故障模式，同时遵守所采样的主题。基于该策略，一个测试代理（文本 LLM）生成初始用户提示文本，该文本通过 TTS 转换为语音，以模拟音频输入。然后，该音频被输入到目标代理（音频 LM）中。规划代理评估目标代理的输出以检测故障。如果模型成功，规划代理将更新策略，在后续回合中引入更高的复杂性。这个循环会重复，直到检测到故障或达到最大回合数。一旦检测到故障，规划代理将分析历史记录 H ，并提取导致模型在该轴上失败的核心逻辑或技巧。这将被综合为一个简洁的人类友好的蓝图 B_{final} ，作为人类贡献者的高级战略指南。在本研究中，我们使用 OpenAI 的 gpt-4o-mini-tts 作为 TTS 系统，并在 GPT 4o 音频预览之间随机采样。

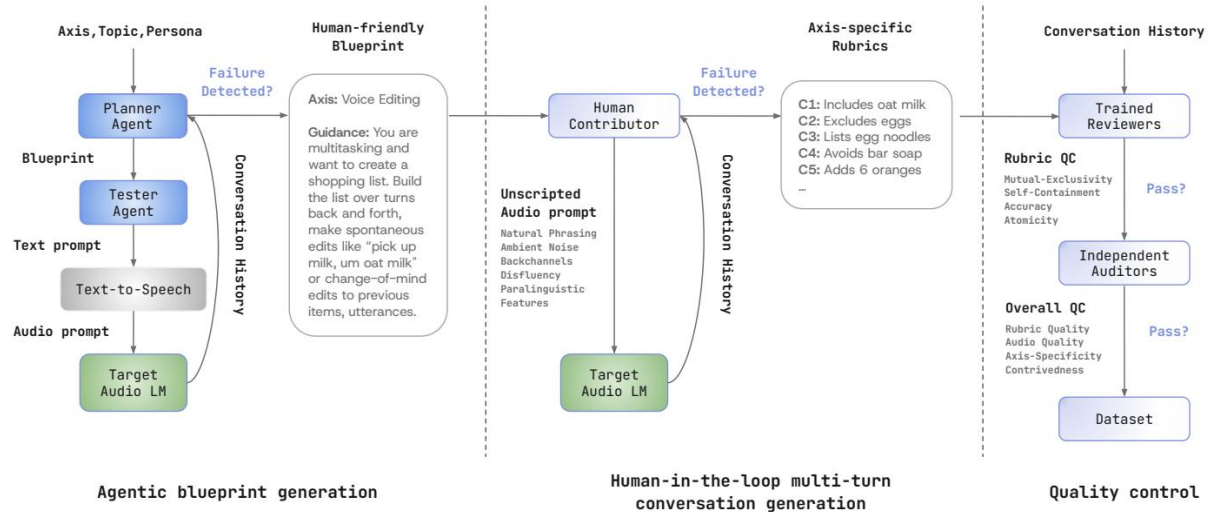


Figure 28: images/81e4ced3d6c9163dcc9c82dd26b5329e90e526a2b1462704519ac5ee634d04f4.jpg

Figure 3: Audio MultiChallenge dataset generation process that utilizes an agentic blueprint generation followed by human-in-the-loop multi-turn conversation generation and quality check. Conversation topic seeds, generated human-friendly blueprints and data samples are provided in the Appendix.

图 3：音频多挑战数据集生成过程，该过程利用了代理蓝图生成，随后进行人类参与的多轮对话生成和质量检查。对话主题种子、生成的人类友好蓝图和数据样本在附录中提供。

and Gemini 2.5 Pro for the Target Agent, and utilize the reasoning capability of OpenAI o3 as the Planner that guides GPT 4o as the User Agent. The full algorithm is provided in Appendix A.6.

用于目标代理的 Gemini 2.5 Pro，并利用 OpenAI o3 的推理能力作为规划器，引导 GPT 4o 作为用户代理。完整的算法见附录 A.6。

Human-In-The-Loop Conversation Generation Using axis-specific blueprints as seeds, we collect audio recordings from human contributors. Each contributor attempts to induce a verifiable failure within 3 to 8 turns. Contributors receive the blueprint and a detailed definition of the target axis, but the blueprint is only strategic guidance. They are instructed to rewrite the plan in their own words, introduce new constraints, or pivot topics as needed to expose model weaknesses. This encourages spontaneous, in-the-wild interaction, including pauses, hesitations, rambling, and mid-utterance self-corrections. On average, 65% of attempted conversations leveraged a synthetic blueprint, highlighting their effectiveness. We provide blueprint samples in Appendix A.2.

使用轴特定的蓝图作为种子进行人参与的对话生成，我们从人类贡献者那里收集音频记录。每位贡献者尝试在 3 到 8 轮内诱导出可验证的失败。贡献者会收到蓝图和目标轴的详细定义，但蓝图仅作为战略指导。他们被指示用自己的话重写计划，引入新的约束条件，或在必要时转换话题以暴露模型的弱点。这鼓励了自发的、真实环境下的互动，包括停顿、犹豫、离题和中途自我更正。平均而言，65% 的尝试对话利用了合成蓝图，突显了它们的有效性。我们在附录 A.2 中提供了蓝图示例。

To collect natural speech data, contributors interact with an Audio LM (either GPT 4o Audio Preview and Gemini 2.5 Pro) in a turn-based manner until a failure is manually detected. For each turn, the contributor records an audio prompt, which is processed by the Target Audio LM. The model output is returned as text for immediate review. After each turn, contributors check that the interaction remains challenging and coherent, avoids premature failures unrelated to the assigned axis, and continues the conversation naturally. If these conditions are met, the contributor proceeds to the next turn. Once an objective failure is triggered under the assigned axis, the

conversation ends.

为了收集自然语音数据，贡献者以回合制的方式与音频语言模型（如 GPT 4o Audio Preview 和 Gemini 2.5 Pro）进行交互，直到手动检测到失败为止。对于每个回合，贡献者录制一个音频提示，该提示会被目标音频语言模型处理。模型输出的结果以文本形式返回，以便立即查看。在每个回合结束后，贡献者检查交互是否保持具有挑战性且连贯，避免与所分配轴无关的过早失败，并自然地继续对话。如果满足这些条件，贡献者将进入下一个回合。一旦在所分配的轴下触发了目标失败，对话将结束。

Human-Authored Rubrics for Model Evaluation Once the conversation is marked as a failure along the specific axis, contributors formulate a set of atomic, binary criteria specifically designed to evaluate the model response of the final turn where the failure occurred. Rather than creating a single, unstructured ideal rubric, we require contributors to decompose the ideal response into a set of multiple discrete rubrics. To ensure these rubrics strictly measure the targeted axes, we instruct contributors to generate criteria that correspond exclusively to both the assigned axis and the final user turn. General conversational requirements related to only the final user turn or generic axis definitions related to the assigned axis only are explicitly excluded. By breaking down the ideal response into self-contained, binary statements, our approach allows an LLM-as-a-judge to verify individual rubric criterion with higher precision than when scoring against a single reference text.

模型评估的人工制定评分标准一旦对话在特定轴线上被标记为失败，贡献者会制定一组原子的、二元的评分标准，专门用于评估导致失败的最后一步中模型的响应。我们不要求贡献者创建一个单一的、无结构的理想评分标准，而是要求他们将理想响应分解为多个独立的评分标准。为了确保这些评分标准严格测量目标轴线，我们指示贡献者生成仅与分配的轴线和最后的用户回合相关的评分标准。仅与最后用户回合相关的一般性对话要求，或仅与分配轴线相关的通用轴线定义，都被明确排除。通过将理想响应分解为独立的、二元的陈述，我们的方法使作为评委的 LLM 能够以比对单一参考文本更高的精度来验证每个评分标准。

Quality Control We construct Audio MultiChallenge with highly trained core contributors and a three-stage review process. Each task is audited by specialized reviewers, an independent quality control team, and our internal research team. We evaluate quality along conversation naturalness, failure validity, prompt diversity, audio quality, and rubric quality, as summarized in Figure 3.

质量控制我们通过高度训练的核心贡献者和三阶段审核流程构建 Audio MultiChallenge。每个任务都会由专门的审核人员、独立的质量控制团队以及我们的内部研究团队进行审核。我们从对话自然度、失败有效性、提示多样性、音频质量和评分标准质量等方面评估质量，如图 3 所示。

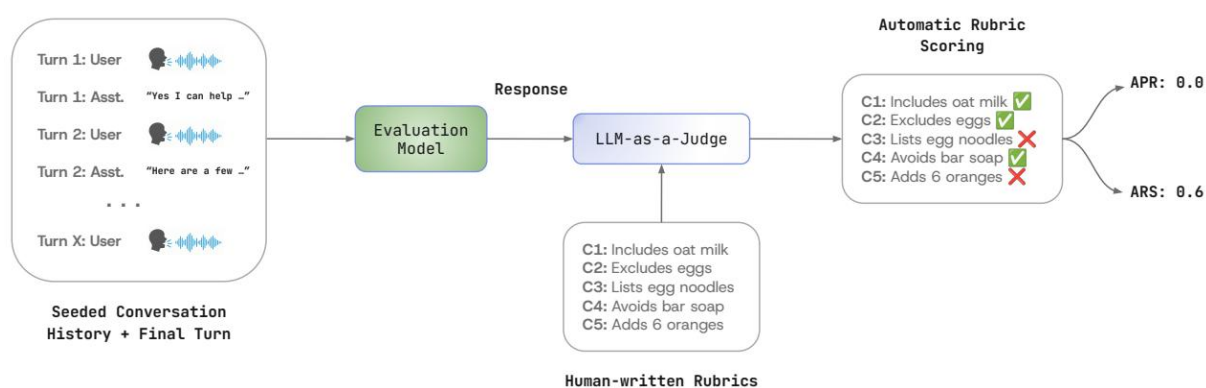


Figure 29: images/8c5e33c737866e7a604abdf33505483b9a084b6a256dd731f0eae3b6b25b091c.jpg

Figure 4: We utilize a fixed-context evaluation protocol where each model receives identical conversation history and is evaluated on its response to the final user turn. An LLM-as-a-judge is subsequently used to evaluate the text stream of the model response against our human-written rubrics.

图 4: 我们采用一种固定上下文的评估协议, 其中每个模型都接收相同的对话历史, 并对其在最后用户回合的回应进行评估。随后, 使用一个大型语言模型作为评判者, 根据我们人工编写的评分标准对模型回应的文本流进行评估。

3. Evaluation

3. 评估

Fixed-Context Evaluation Protocol To ensure a standardized evaluation, we adopt a fixed-context protocol, as depicted in Figure 4. Specifically, for each test example, we pre-seed the full conversation history (both user and assistant turns) and evaluate only the model’s response to the final turn using our LLM-as-a-judge approach. User turns are provided in context as base64-encoded audio files, while assistant turns are provided as raw text. This contrasts with sequential evaluation methods used in prior work, where a model generates responses turn-by-turn to pre-recorded user prompts [20]. We avoid the sequential approach because it can suffer from conversation drift: if a model’s response in the first turn diverges even slightly from the reference transcript, subsequent pre-recorded user follow-ups and their associated rubrics may become semantically invalid. Our approach encapsulates more natural back-and-forth interaction patterns and ensures that every model receives an identical context state, enabling a fair, reproducible comparison.

固定上下文评估协议为了确保评估的标准化, 我们采用固定上下文协议, 如图 4 所示。具体来说, 对于每个测试示例, 我们预先设置完整的对话历史记录 (包括用户和助手的发言), 并仅使用我们的 LLM-as-a-judge 方法评估模型对最后一轮的回应。用户发言以 base64 编码的音频文件形式提供上下文, 而助手发言则以原始文本形式提供。这与之前工作中使用的顺序评估方法不同, 其中模型会逐轮生成响应以应对预录制的用户提示 [20]。我们避免使用顺序方法, 因为它可能导致对话偏离: 如果模型在第一轮的回应与参考脚本略有偏差, 后续预录制的用户跟进及其相关的评分标准可能会变得语义无效。我们的方法包含了更自然的来回互动模式, 并确保每个模型都接收到相同的上下文状态, 从而实现公平且可重复的比较。

Evaluation Metrics We utilize the Average Pass Rate (APR) metric [23] to evaluate model performance on Audio MultiChallenge. Concretely, a model’s response is considered successful if and only if it satisfies all corresponding rubrics; otherwise, it is marked as a failure. We report the Overall APR as the percentage of tasks successfully completed across the dataset. This methodology aligns with MultiChallenge [17], where a single, binary weighted target question was defined to grade each task. Instead of restricting our data to one rubric criteria, we allow for multiple binary weighted atomic rubrics for increased LLM judge reliability, and equivalently calculate model performance as their APR. Additionally, we report an Average Rubric Score (ARS) [23] or a rubric compliance [24] score for each model which represents the average percentage of satisfied rubrics per task. We define our evaluation metrics formally in Appendix A.4.

评估指标我们使用平均通过率 (APR) 指标 [23] 来评估模型在 Audio MultiChallenge 上的性能。具体来说, 只有当模型的响应满足所有相应的评分标准时, 才认为该响应是成功的; 否则, 将其标记为失败。我们报告总体的 APR 作为数据集中成功完成任务的百分比。这种方法与 MultiChallenge [17] 一致, 其中定义了一个单一的加权目标问题来评估每个任务。与限制我们的数据仅使用一个评分标准不同, 我们允许使用多个加权的原子评分标准, 以提高 LLM 判定的可靠性, 并等效地将模型性能计算为它们的 APR。此外, 我们还报告每个模型的平均评分标准分数 (ARS) [23] 或评分标准符合度 [24] 分数, 该分数表示每个任务中满足评分标准的平均百分比。我们在附录 A.4 中正式定义了我们的评估指标。

3.1 Results

3.1 结果

We present scores on Audio MultiChallenge across a variety of E2E architectures in Table 2, along with their performance per-axis. We observe that Gemini 3 Pro Preview (Thinking) achieves the highest overall score of 54.65%, followed by its counterparts Gemini 2.5 Pro (Thinking) and 2.5

Flash (Thinking). In comparison, the remainder of proprietary and open-source models perform modestly, with GPT 4o Audio Preview and Voxtral Small 24B leading, scoring a 25.44% and 26.33% APR respectively, closely followed by GPT Realtime with an APR of 23.45% (output modality set to Text). Open-source S2S architectures in particular struggle the most, with no model surpassing Qwen 3 Omni which achieves a modest 24.34% APR. Analyzing per-axis scores, our newly adapted Voice Editing and Inference Memory prove to be the most challenging tasks for frontier models, with them scoring 17.99% and 21.55% on average, respectively. We highlight the following key findings and provide samples of model failures in Appendix A.1.

我们在表 2 中展示了在多种端到端 (E2E) 架构下在 Audio MultiChallenge 上的得分，以及它们在各个轴上的表现。我们观察到，Gemini 3 Pro Preview (Thinking) 取得了最高的总体得分 54.65%，其次是其同类模型 Gemini 2.5 Pro (Thinking) 和 2.5 Flash (Thinking)。相比之下，其余专有和开源模型的表现较为平庸，其中 GPT 4o Audio Preview 和 Voxtral Small 24B 领先，分别获得了 25.44% 和 26.33% 的 APR，紧随其后的是 GPT Realtime，其 APR 为 23.45%（输出模式设置为文本）。特别是开源的 S2S 架构在表现上最为困难，没有任何模型能超过 Qwen 3 Omni，其 APR 仅为 24.34%。分析各个轴的得分，我们新适应的语音编辑和推理记忆被证明是前沿模型中最困难的任务，分别平均得分为 17.99% 和 21.55%。我们在附录 A.1 中突出了以下关键发现，并提供了模型失败的示例。

Model	Output Modality	Overall		Per-Axis APR (%)			
		IM	IR	SC	VE		
APR (%)	ARS (%)						
Gemini 3 Pro Preview (Thinking)	Text	54.65	82.54	57.58	65.283	40.96	49.57
Gemini 2.5 Pro (Thinking) [25]	Text	46.90	78.91	44.70	56.67	40.96	43.59
Gemini 2.5 Flash (Thinking)	Text	40.04	74.59	30.30	37.50	54.22	43.59
Gemini 2.5 Flash	Text	26.11	65.42	19.70	29.17	31.33	26.50
GPT Realtime 1	Text	23.45	65.95	21.97	26.67	22.89	22.22
GPT Realtime	Audio	20.35	63.03	19.70	20.00	26.51	17.09
GPT 4o Audio Preview [26]	Text	25.44	67.58	23.48	27.50	31.33	21.37
GPT 4o Audio Preview	Audio	23.23	64.25	21.21	30.83	24.10	17.09
GPT 4o Mini Audio Preview	Text	14.82	57.01	15.15	16.67	20.48	8.55

GPT 4o Mini Audio Preview	Audio	13.05	54.91	9.85	12.50	21.69	11.11
Qwen 3 Omni [7]	Audio	24.34	63.43	20.45	27.50	30.12	21.37
Qwen 2.5 Omni [27]	Text	11.95	45.39	12.88	13.33	18.07	5.13
Voxtral Small 24B [28]	Text	26.33	66.59	20.45	25.00	27.71	33.33
Mimo Audio 7B Instruct (Thinking) [29] ¹	Text	19.69	55.78	19.70	22.50	27.71	11.11
Mimo Audio 7B Instruct	Text	18.58	53.86	17.42	20.00	27.71	11.97
Phi 4 MM Instruct [30]	Text	15.49	46.96	16.67	16.67	28.92	3.42
Gemma 3n E4B IT [31]	Text	15.49	43.11	14.39	22.50	26.51	1.71
Kimi Audio 7B Instruct [32]	Text	13.72	48.36	14.39	16.67	20.48	5.13
Kimi Audio 7B Instruct	Audio	10.40	22.08	15.91	6.67	13.25	5.98
LFM2 Audio 1.5B [33]	Audio	9.29	19.10	15.15	7.50	15.66	0.00

模型	输出模态	总体上	按轴的 APR (%)				
APR (%)	ARS (%)	IM	IR	SC	VE		
Gemini 3 Pro 预览 (思考)	文本	54.65	82.54	57.58	65.283	40.96	49.57
Gemini 2.5 Pro (思考中) [25]	Text	46.90	78.91	44.70	56.67	40.96	43.59
Gemini 2.5 Flash (思考)	Text	40.04	74.59	30.30	37.50	54.22	43.59
Gemini 2.5 Flash	Text	26.11	65.42	19.70	29.17	31.33	26.50
GPT 实时 ¹	Text	23.45	65.95	21.97	26.67	22.89	22.22
GPT 实时	音频	20.35	63.03	19.70	20.00	26.51	17.09
GPT 4o Audio Preview [26]	文本	25.44	67.58	23.48	27.50	31.33	21.37
GPT 4o 音频预览	音频	23.23	64.25	21.21	30.83	24.10	17.09
GPT 4o Mini 音频预览	Text	14.82	57.01	15.15	16.67	20.48	8.55

GPT 4o Mini 音频预览	音频	13.05	54.91	9.85	12.50	21.69	11.11
Qwen 3 Omni [7]	音频	24.34	63.43	20.45	27.50	30.12	21.37
Qwen 2.5 Omni [27]	文本	11.95	45.39	12.88	13.33	18.07	5.13
Voxtral Small 24B [28]	Text	26.33	66.59	20.45	25.00	27.71	33.33
Mimo Audio 7B Instruct (思考中) [29] ¹	Text	19.69	55.78	19.70	22.50	27.71	11.11
Mimo Audio 7B Instruct	文本	18.58	53.86	17.42	20.00	27.71	11.97
Phi 4 MM Instruct [30]	Text	15.49	46.96	16.67	16.67	28.92	3.42
Gemma 3n E4B IT [31]	文本	15.49	43.11	14.39	22.50	26.51	1.71
Kimi Audio 7B Instruct [32]	文本	13.72	48.36	14.39	16.67	20.48	5.13
Kimi Audio 7B Instruct	音频	10.40	22.08	15.91	6.67	13.25	5.98
LFM2 Audio 1.5B [33]	音频	9.29	19.10	15.15	7.50	15.66	0.00

Table 2: Overall and Per-Axis scores on Audio MultiChallenge per model and output modality. APR serves as final scores. IM: Inference Memory, IR: Instruction Retention, SC: Self Coherence, VE: Voice Editing. Green highlights indicate best performance with output modality as Text; blue highlights indicate best Audio output performance.

表 2: 各模型和输出模态在 Audio MultiChallenge 上的总体得分和每轴得分。APR 作为最终得分。IM: 推理记忆, IR: 指令保持, SC: 自我连贯性, VE: 语音编辑。绿色高亮表示在文本输出模态下的最佳表现; 蓝色高亮表示在音频输出模态下的最佳表现。

Performance varies over output modalities We observe a “modality gap” when evaluating models that support both text and audio output streams. Under a Text only output configuration, these models score 19.36% on average, however, when the same models are configured to output Audio (or joint Audio and Text depending on their architecture), their performance degrades to 16.76%. These findings are independently reported in other recent works [29], highlighting the need for improved S2S post-training data to overcome this gap in performance.

性能在输出模态之间有所不同。我们在评估支持文本和音频输出流的模型时，观察到一个“模态差距”(modality gap)。在仅文本输出的配置下，这些模型的平均得分是 19.36%，然而，当这些模型配置为输出音频（或根据其架构输出音频和文本）时，其性能下降到 16.76%。这些发现已在其他近期的研究中独立报告 [29]，突显了需要改进的端到端（S2S）后训练数据以克服这一性能差距。

Audio-Cue Inference Memory lags substantially We classify Audio-Cue gated Inference Memory tasks ($n = 42$) as ones where the information the model must remember from a previous turn is contained only in the audio signal rather than in the words spoken, as described in Section 2.1. Our examples span both paralinguistic cues such as the speaker’s tone and ambient cues such as background sound. Figure 5 shows a significant gap in remembering these audio cues in our multi-turn setup. Across most models, performance drops substantially when the task requires inferring paralinguistic features or ambient sounds from previous turns, with ARS falling by 36.5% relatively compared to semantic memory on average across models. This unique subset of our data highlights the limitations of current E2E spoken dialogue systems when it comes to audio-native state tracking and recall, which is fundamental to realtime and multi-turn voice conversations.

音频线索推理记忆存在显著滞后我们将音频线索门控推理记忆任务 ($n = 42$) 归类为一类任务，其中模型需要从之前的对话回合中记忆的信息仅包含在音频信号中，而非口头表达的内容，如第 2.1 节所述。我们的示例涵盖了诸如说话者的语气等副语言线索以及背景声音等环境线索。图 5 显示了在我们的多轮对话设置中，记住这些音频线索存在显著差距。在大多数模型中，当任务需要从之前的对话回合中推断副语言特征或环境声音时，性能会显著下降，平均而言，与语义记忆相比，ARS 下降了 36.5%。这一独特的数据集突显了当前端到端语音对话系统在音频原生状态跟踪和回忆方面的局限性，而这对于实时和多轮语音对话至关重要。

Conditional Instructions cause model failures Frontier models score 25.08% APR and 46.11% ARS on average on Instruction Retention (IR) tasks. To analyze the main sources of error, we automatically cluster each IR rubric item into two instruction types: Fixed, which are unconditional, persistent constraints meant to hold throughout the conversation (e.g., response style or roleplay prompts), and Conditional, which are context-dependent instructions that only apply in specific future-turn scenarios. Figure 6 shows that each model, regardless of their overall performance, demonstrates worse performance in conditional IR. These results support treating conditional instruction following as a distinct, harder sub-problem within IR, and they motivate future work on reliably retrieving and applying conditional rules rather than optimizing only for conventional instruction following.

条件指令导致模型失效。前沿模型在指令保留（IR）任务上的平均年化收益率（APR）为 25.08%，平均回报率（ARS）为 46.11%。为了分析错误的主要来源，我们将每个 IR 评分标准项目自动聚类为两种指令类型：固定指令，即无条件、持久的约束，旨在在整个对话中保持有效（例如，响应风格或角色扮演提示），以及条件指令，即依赖于上下文的指令，仅在特定未来回合场景中适用。图 6 显示，无论模型的整体表现如何，它们在条件 IR 任务上的表现都较差。这些结果支持将条件指令遵循视为 IR 中的一个独立、更困难的子问题，并激励未来的工作重点在于可靠地检索和应用条件规则，而不仅仅是优化传统的指令遵循。

Self Coherence linearly declines on tasks with increased context Since the models we benchmark use different audio tokenizers, raw token counts are not directly comparable across systems. We therefore use cumulative

自我一致性在上下文增多的任务中线性下降。由于我们基准测试的模型使用了不同的音频分词器，因此原始分词数量在不同系统之间无法直接比较。因此，我们使用累计

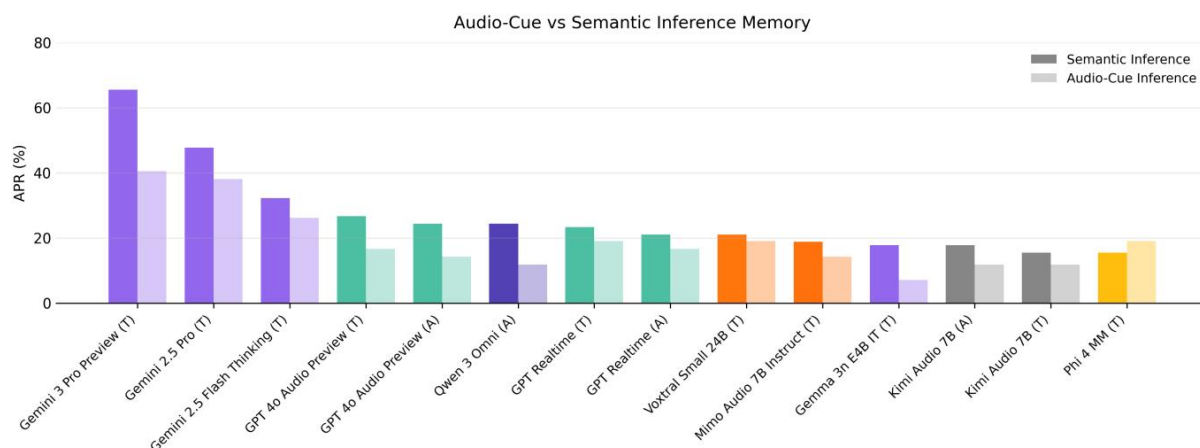


Figure 30: images/3f5bb6153e3d695c567a63ce0511ec3dca30a6c80f9bbfe2fbadad6279d4d0cd.jpg

Figure 5: APR (%) scores on Audio-Cue vs. Semantic Inference Memory tasks. (T) indicates text output, (A) indicates audio output.

图 5：音频线索与语义推理记忆任务中的 APR（%）得分。（T）表示文本输出，（A）表示音频输出。

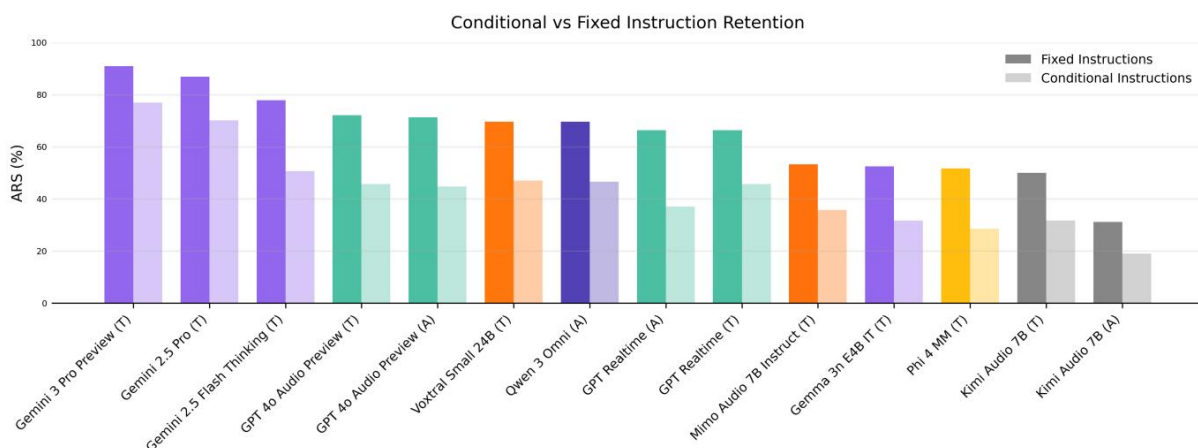


Figure 31: images/d6f067a29331cc1fc81574c75a299059c5e06b744059094b909d5db7b1c1587d.jpg

Figure 6: ARS (%) on Instruction Retention tasks classified by rubric conditionality. ARS is reported instead of APR since the classification is at the rubric level – whereas a single task can have both conditional and fixed rubrics. (T) indicates text output, (A) indicates audio output.

图 6：按评分细则条件性分类的指令保留任务的 ARS (%)。由于分类是在评分细则层面进行的，因此使用 ARS 而非 APR，因为一个任务可能同时包含条件性评分细则和固定评分细则。(T) 表示文本输出，(A) 表示音频输出。

user-audio duration per task as a proxy for audio context length. Self Coherence degrades sharply as cumulative user audio increases. Figure 8b shows that APR on Self Coherence drops from 33.3% for tasks with 0–60 seconds of cumulative user audio to 20.0% for tasks with 3–5 minutes, and further to 13.3% for tasks exceeding 5 minutes (though this final bin contains only 3 samples) when averaged across all models. We analyze duration effects on other axes in Section 3.2.

以每个任务的用户音频时长作为音频上下文长度的代理。随着累积用户音频的增加，自我连贯性急剧下降。图 8b 显示，在所有模型平均的情况下，对于累积用户音频时长为 0–60 秒的任务，自我连贯性的 APR 为 33.3%，而对于时长为 3–5 分钟的任务，APR 下降至 20.0%，而对于超过 5 分钟的任务，APR 进一步降至 13.3%（尽管最后一个区间仅包含 3 个样本）。我们在第 3.2 节中分析了时长对其他维度的影响。

Mid-Utterance and Prior-Turn Voice Edits hurt performance Voice Editing remains our most challenging axis, with an average APR of 17.99%. Most failures on Voice Editing occur when users introduce multiple mid-utterance (in-turn speech repairs) and prior-turn edits throughout a conversation, with the final prompt typically requesting a summary or revision that integrates all previously specified changes. Successfully completing these tasks requires strong state tracking and multi-constraint integration at the semantic level, as well as robustness to user backchannels and disfluencies at the acoustic level. This helps explain why stronger or reasoning-heavy models can be similarly robust to mid-utterance and prior-turn edits and to unedited rubric items corresponding to simple instructions that were never modified, as shown in Figure 7. Our ablation studies in Section 3.4 further highlight the difficulty of this axis.

中句和前轮语音编辑会损害性能语音编辑仍然是我们最具挑战性的轴线，平均 APR 为 17.99%。大多数语音编辑失败发生在用户在整个对话中引入多个中句（同轮语音修正）和前轮编辑时，最终提示通常要求总结或修订，整合所有先前指定的更改。成功完成这些任务需要在语义层面具备强大的状态跟踪和多约束整合能力，以及在声学层面对用户背景音和不流畅部分的鲁棒性。这解释了为什么更强或推理密集型模型可以同样应对中句和前轮编辑，以及未编辑的评分项目，这些项目对应于从未被修改的简单指令，如图 7 所示。我们在第 3.4 节中的消融研究进一步突显了这一轴线的难度。

Rubric Satisfaction Rate by Voice Edit Type

按语音编辑类型划分的评分满意度率

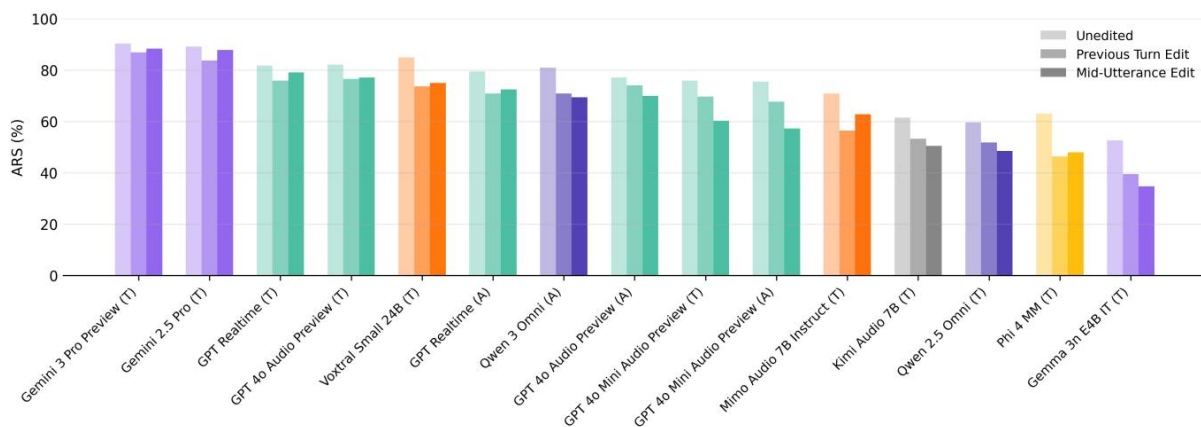


Figure 32: images/3a35ec837f29c3daf825442f0b341010df405a4a3da74143f5db5b5cc4304525.jpg

Figure 7: ARS (%) on Voice Editing tasks classified by the edit type made by the user for each rubric. Unedited refers to statements that the user did not backtrack or revise either within the same turn or in future turns. (T) indicates text output, (A) indicates audio output.

图 7: 按用户在每个评分标准中所进行的编辑类型对语音编辑任务的 ARS (%) 进行分类。未编辑 (Unedited) 指的是用户在当前回合或未来回合中没有回退或修改的陈述。(T) 表示文本输出, (A) 表示音频输出。

3.2 Analysis

3.2 分析

Turn Count Figure 8a shows no meaningful relationship between turn count and performance: the scores in Audio MultiChallenge remain largely stable as the number of turns increases. Although the benchmark is multi-turn, the conversations are short relative to the context windows of modern models, so adding turns does not mechanically increase difficulty. This mirrors findings from prior work using fixed-context evaluation protocols [17]. Since each turn's length is also arbitrarily defined, it is a weak measure for conversation histories that models must parse through. Furthermore, since the history for all previous turns is provided to the model and is audited (i.e., free of assistant hallucinations or errors), models can often leverage additional in-context examples at higher turn counts to respond consistently. We observe similarly weak trends when analyzing performance by axis.

转数与表现的关系图 8a 显示了转数与表现之间没有有意的关系：随着转数的增加，音频多挑战任务的得分保持相对稳定。尽管基准测试是多轮的，但与现代模型的上下文窗口相比，对话长度较短，因此增加转数并不会机械地提高难度。这之前使用固定上下文评估协议的研究结果一致 [17]。此外，由于每一轮的长度也是任意定义的，因此它作为模型需要解析的对话历史的衡量标准是不充分的。此外，由于模型会收到所有先前轮次的历史信息并进行审核（即没有助手的幻觉或错误），因此模型通常可以在更高的转数下利用额外的上下文示例来保持一致的响应。当我们按轴分析表现时，我们观察到类似弱的趋势。

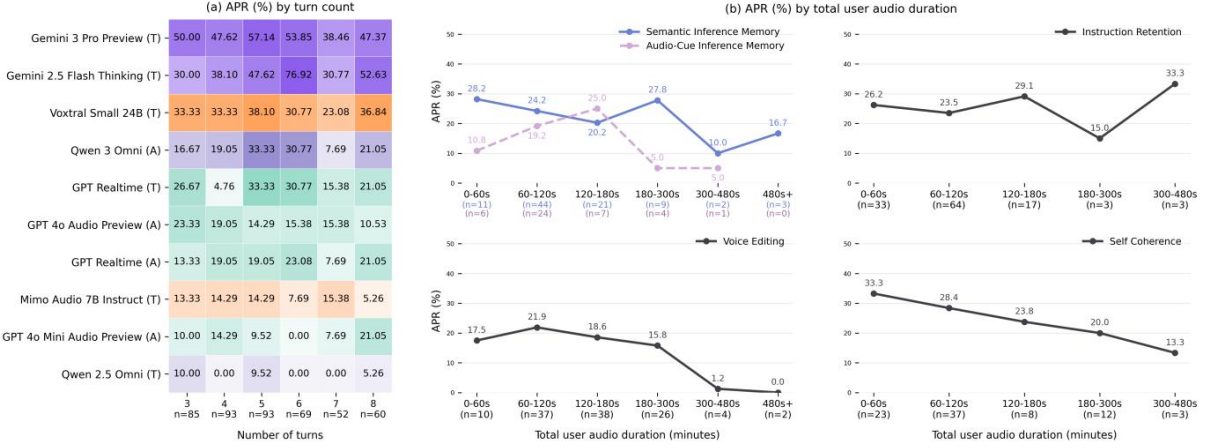


Figure 8: APR (%) across (a) number of turns, (b) binned total user audio duration (in seconds)

图 8: APR (%) 在 (a) 转数数量, (b) 按总用户音频时长 (秒) 分组的统计情况

Audio Duration Unlike turn count, total user-audio duration is a more direct proxy for effective context length. Figure 8b shows a clear degradation in Self Coherence as audio duration increases, consistent with the trend discussed above. Voice Editing also worsens with longer histories as average APR falls from 17.5% in the 3–5 minute bin to 1.2% and 0% for 5–8 minute and 8+ minute tasks respectively (though these bins have low sample sizes $n = 4, 2$). Inference Memory declines more modestly with duration, driven primarily by Semantic Inference Memory, where longer contexts make targeted recall harder. By contrast, Audio-Cue Inference Memory shows no clear trend with duration; qualitatively, we observe performance is often bottlenecked first by audio perception

音频时长与回合数不同，总用户音频时长是更直接的有效上下文长度的代理指标。图 8b 显示，随着音频时长的增加，自我连贯性出现了明显的下降，这与上述讨论的趋势一致。语音编辑效果也随着更长的历史记录而变差，因为在 3-5 分钟区间内平均 APR 为 17.5%，而在 5-8 分钟和 8 分钟以上的任务中分别下降至 1.2% 和 0%（尽管这些区间的样本量较小， $n=4, 2$ ）。推理记忆随着时长的增加而更为适度地下降，这主要由语义推理记忆驱动，其中更长的上下文使得有针对性的回忆变得更加困难。相比之下，音频线索推理记忆在时长方面没有明显的趋势；从定性上看，我们观察到性能通常首先受到音频感知的限制

(e.g., reliably identifying paralinguistic cues or background sounds) rather than retention per se, leading to failures even on short clips. Interestingly, Instruction Retention remains steady or even improves on longer audio tasks because many instruction-following tasks introduce conditional or fixed constraints early in the dialogue that must be followed throughout. Under our fixed-context protocol, the model also receives audited prior assistant turns that already follow these constraints, providing additional in-context demonstrations it can imitate in later turns, consistent with in-context learning effects observed on instruction following in LLMs [34].

（例如，可靠地识别副语言线索或背景声音）而不是记忆本身，导致即使在短片段上也会出现失败。有趣的是，指令保留（Instruction Retention）在更长的音频任务中保持稳定甚至有所提高，因为许多遵循指令的任务在对话早期就引入了条件性或固定的约束，这些约束在整个对话过程中都必须遵循。在我们的固定上下文协议下，模型还会接收到经过审核的先前助手回合，这些回合已经遵循了这些约束，从而为模型在后续回合中提供额外的上下文示例，这与在大型语言模型（LLMs）中观察到的上下文学习效应一致 [34]。

Response Length One hypothesis for why models score higher on rubric-based benchmarks is their scores may correlate with response length [24]. Since rubrics evaluate against an explicit list of criteria, an overly verbose model could satisfy more rubric items even if the output is not realistic or human-desirable. A common strategy to mitigate this is to include negatively weighted criteria for incorrect or undesirable content. For Audio MultiChallenge, we adapt this approach by utilizing exclusion rubrics (e.g., “Avoids mentioning leather products”) that explicitly check whether the model omits information the user instructed not to mention, later edited

响应长度一个关于为何模型在基于评分标准的基准测试中得分较高的假设是，其得分可能与响应长度相关

[24]。由于评分标准是根据明确的评估标准进行评估的，一个过于冗长的模型可能会满足更多的评分标准项目，即使输出内容并不现实或符合人类期望。一种常见的缓解策略是加入对错误或不受欢迎内容的负权重评估标准。对于 Audio MultiChallenge，我们采用这种方法，通过使用排除性评分标准（例如，“避免提及皮革产品”）来明确检查模型是否遗漏了用户要求不提及的信息，之后进行编辑。

away, or signaled a preference to avoid. Figure 9 plots rubric compliance (ARS) against each model’s average response length. Response length is measured by tokenizing each model’s text output stream with the tiktoken library. Despite these controls, we observe a strong, approximately linear relationship between response length and ARS. To assess whether this reflects score hacking, we independently review responses from the Gemini family and find no evidence that their higher scores are driven primarily by verbosity. Instead, their outputs more consistently reflect stronger audio and semantic understanding across axes and more reliable state tracking.

远离，或表明了避免的偏好。图 9 将评分标准符合度（ARS）与每个模型的平均响应长度进行对比。响应长度通过使用 tiktoken 库对每个模型的文本输出流进行分词来测量。尽管采取了这些控制措施，我们仍观察到响应长度与 ARS 之间存在强烈的、近似线性的关系。为了评估这种现象是否反映了评分操纵，我们独立审查了 Gemini 系列模型的响应，并未发现其较高分数主要由冗长性驱动的证据。相反，其输出在各个维度上更一致地反映了更强的音频和语义理解能力，以及更可靠的语境跟踪能力。

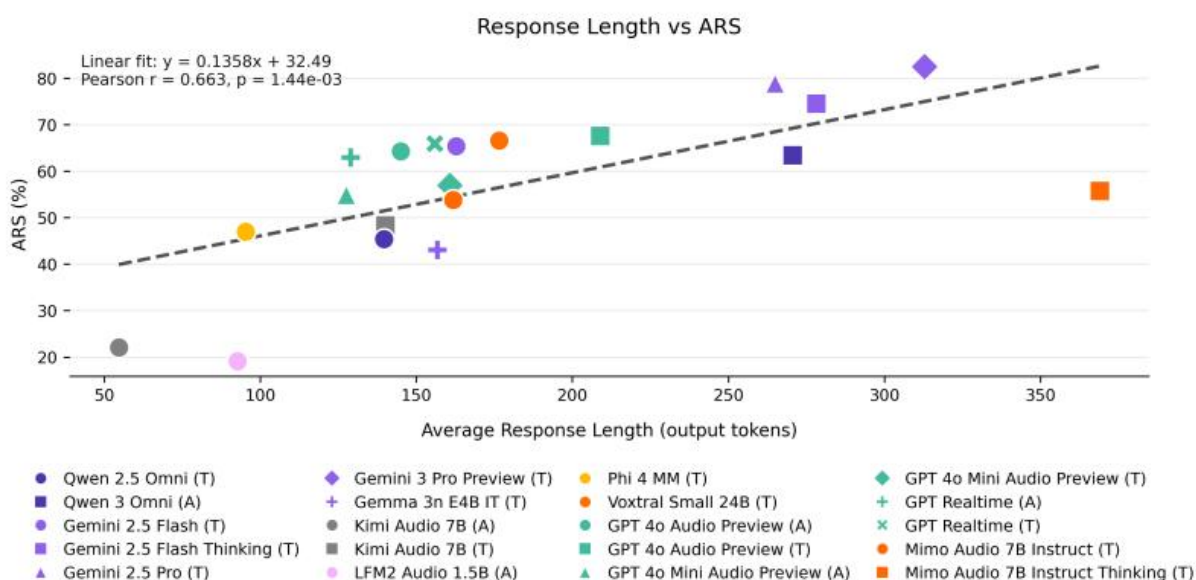


Figure 33: images/a503addaa1bf0a071116ace33bf44cf2df8243f5f9c4a943229002c2b554b313.jpg

Figure 9: Rubric compliance (ARS) as a function of the model’s average response length in tokens.

图 9：评分标准符合度（ARS）作为模型平均响应长度（以 token 为单位）的函数。

3.3 LLM-as-a-Judge

3.3 LLM-as-a-Judge

To evaluate our LLM-as-a-judge rubric-grading setup, we measure inter-rater agreement between model-generated and human-assigned labels across the full dataset. Human annotators are instructed to grade randomly sampled GPT 4o Audio Preview or Gemini 2.5 Pro responses according to their corresponding written rubrics.

为了评估我们的 LLM 作为法官的评分体系，我们测量模型生成的标签与人类分配的标签在完整数据集

上的评分者间一致性。人类标注员被指示根据相应的书面评分标准，对随机抽取的 GPT 4o 音频预览或 Gemini 2.5 Pro 回复进行评分。

We aggregate these rubric grades into a set of 1,712 observations and compute Cohen’s κ and Macro F1 scores calculated following Arora et al. [35]. To mitigate the self-enhancement bias known to affect LLM judges [36, 37], we exclude our evaluated ALMs which are capable of text-based prompts from judging. Our results, presented in Table 3, show that all

我们将这些评分标准等级汇总为一组 1,712 个观测值，并计算 Cohen’s κ 和按照 Arora 等人 [35] 的方法计算的 Macro F1 分数。为了减轻已知会影响 LLM 评委的自我增强偏见 [36, 37]，我们排除了那些能够基于文本提示进行判断的被评估的 ALMs。我们的结果如表 3 所示，表明所有

judges achieve extremely high agreement with human graders, indicating the effectiveness of our rubric creation and overall LLM-as-a-Judge setup. Among the tested judges, o4 Mini attains the highest performance, with a Macro F1 score of 0.937, and is therefore selected as our primary LLM judge. We provide our LLM-as-a-Judge setup in Appendix A.3.

法官们与人类评分者达成极高的共识，这表明我们的评分标准制定和整体的 LLM 作为法官的设置是有效的。在测试的法官中，o4 Mini 表现最佳，其 Macro F1 得分为 0.937，因此被选为我们主要的 LLM 法官。我们在附录 A.3 中提供了我们的 LLM 作为法官的设置。

Judge	Cohen’s κ	Macro F1
o4 Mini	0.873	0.937
GPT 5 Mini	0.870	0.935
GPT 4.1	0.811	0.906
Claude Haiku 4.5	0.808	0.904
Mistral Medium	0.787	0.894
DeepSeek V3.1	0.765	0.882

法官	Cohen’s κ	宏观 F1
o4 Mini	0.873	0.937
GPT 5 Mini	0.870	0.935
GPT 4.1	0.811	0.906
Claude Haiku 4.5	0.808	0.904
Mistral Medium	0.787	0.894
DeepSeek V3.1	0.765	0.882

Table 3: LLM Judge agreement with humans.

表 3: LLM 法官与人类的一致性。

3.4 Ablation Studies

3.4 消融实验

TTS vs. Human User Audio To isolate the impact of spontaneous and disfluent speech, we evaluate our models on synthetic versions of our human user audio, generated by first performing ASR using whisper-large-v3 followed by TTS using gpt-4o-mini-tts. We exclude Audio-Cue IM and tasks

that contain interruptions or sidebar conversations, resulting in a filtered subset of 385 samples for this study. Results in Table 4a depict models

TTS 与人类用户音频相比，为了隔离自发性和不流畅语音的影响，我们在由以下过程生成的合成版本的人类用户音频上评估我们的模型：首先使用 whisper-large-v3 进行 ASR，然后使用 gpt-4o-mini-tts 进行 TTS。我们排除了 Audio-Cue IM 以及包含中断或侧边栏对话的任务，从而得到用于本研究的过滤后的子集样本共 385 个。表 4a 中的结果显示了模型的性能。